LAB TASK 1

COMPUTER NETWORKS

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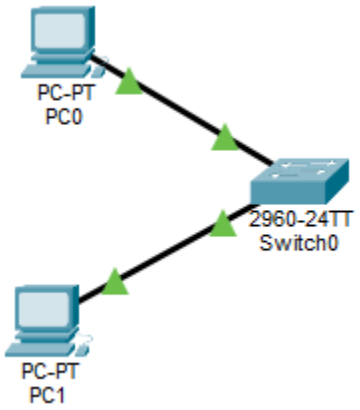
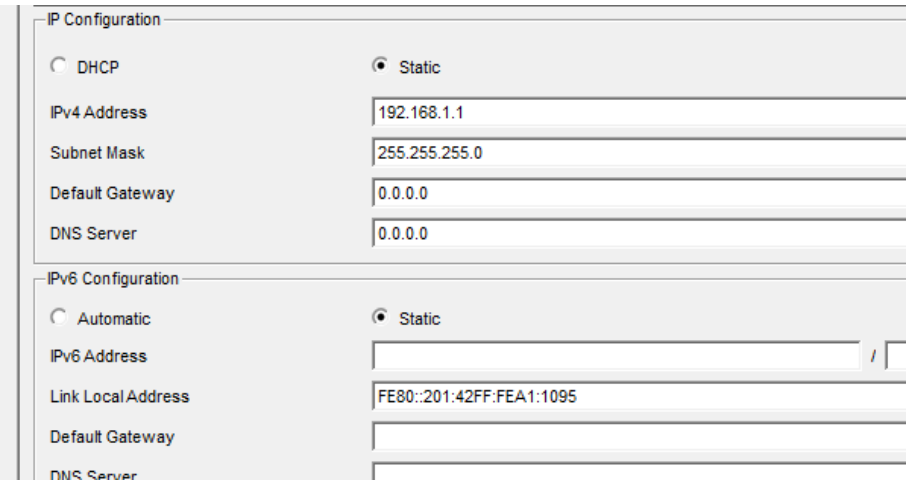
TASK: Create and test a simple LAN in cisco packet tracer

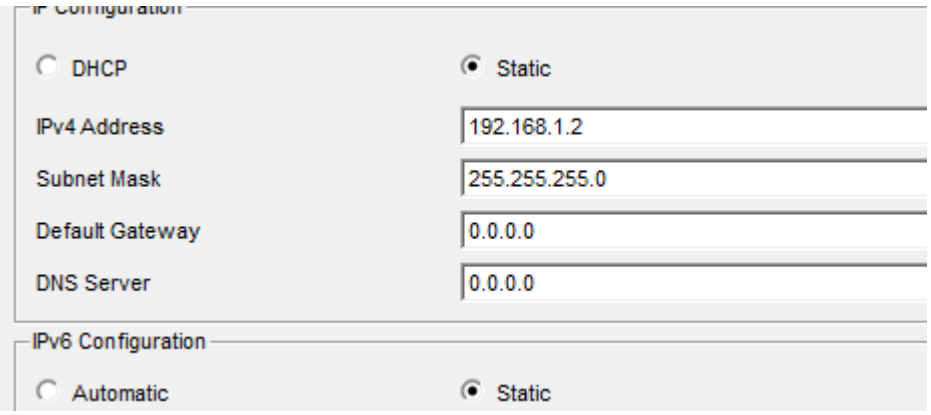
Step 1:



In this step we opened cisco packet tracer and created a simple LAN .one switch and two PCs were added to the workspace. Both PCs were connected to the switch using copper straight cables.

Step 2:





In this step we configured IP addresses manually on both PCs .Pc0 was assigned the IP addresses 192.168.1.1 and PC1 was assigned the IP address 192.168.1.2 with subnet mask 255.255.255.0.

Step 3:

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:42FF:FEA1:1095
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.1
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::290:2BFF:FEB5:81AE
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
```

In this step the ipconfig command was used in the command prompt of each PC to display the IP configuration. This command show the IP address , subnet mask and other network details of the device.

Step 4:

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

In this step we used ping command that is used to test the connectivity between the two PCs PC0 pinged the IP address of PC1 and PC1 pinged the IP address of PC0. The successful “reply from” messages confirmed that both PCs were connected and communicating properly.

Step 5:

```
C:\>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.2

Trace complete.
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 192.168.1.1

Tracing route to 192.168.1.1 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.1

Trace complete.
```

The tracert command shows the path taken by data packets from one device to another. In this network , both PCs are connected through a one switch so the result shows only one hop.

Step 6:

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.2           0090.2bb5.81ae        dynamic

C:\>
```

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0001.42a1.1095        dynamic
```

The arp –a command display the ARP table which mapa IP addresses to MAC addresses.

After sending the ping this command show the MAC address to the other PC.